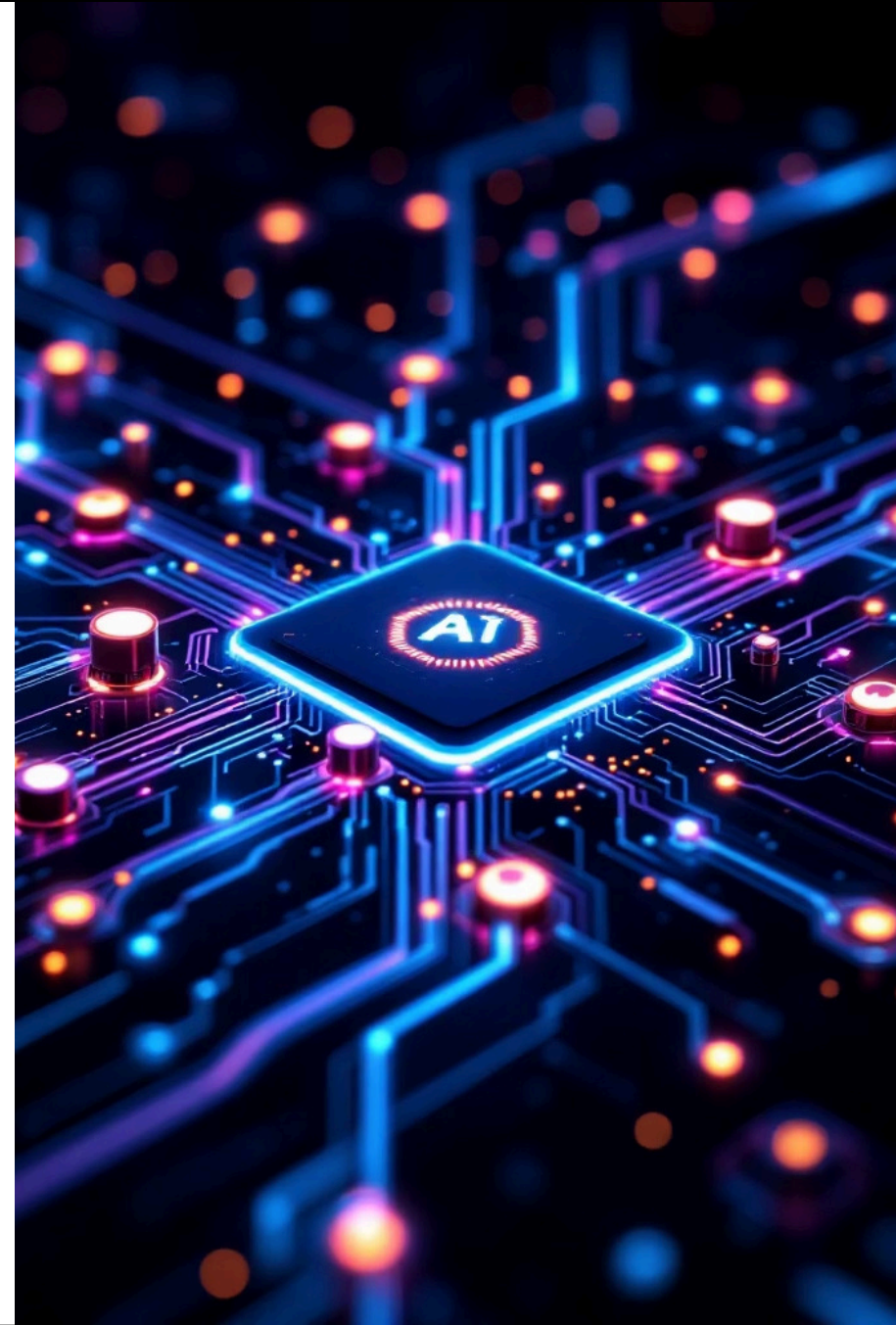


RUFLO v3.5 — Technical Architecture Deep Dive

Multi-Agent Orchestration at Scale | By RuvNet Engineering

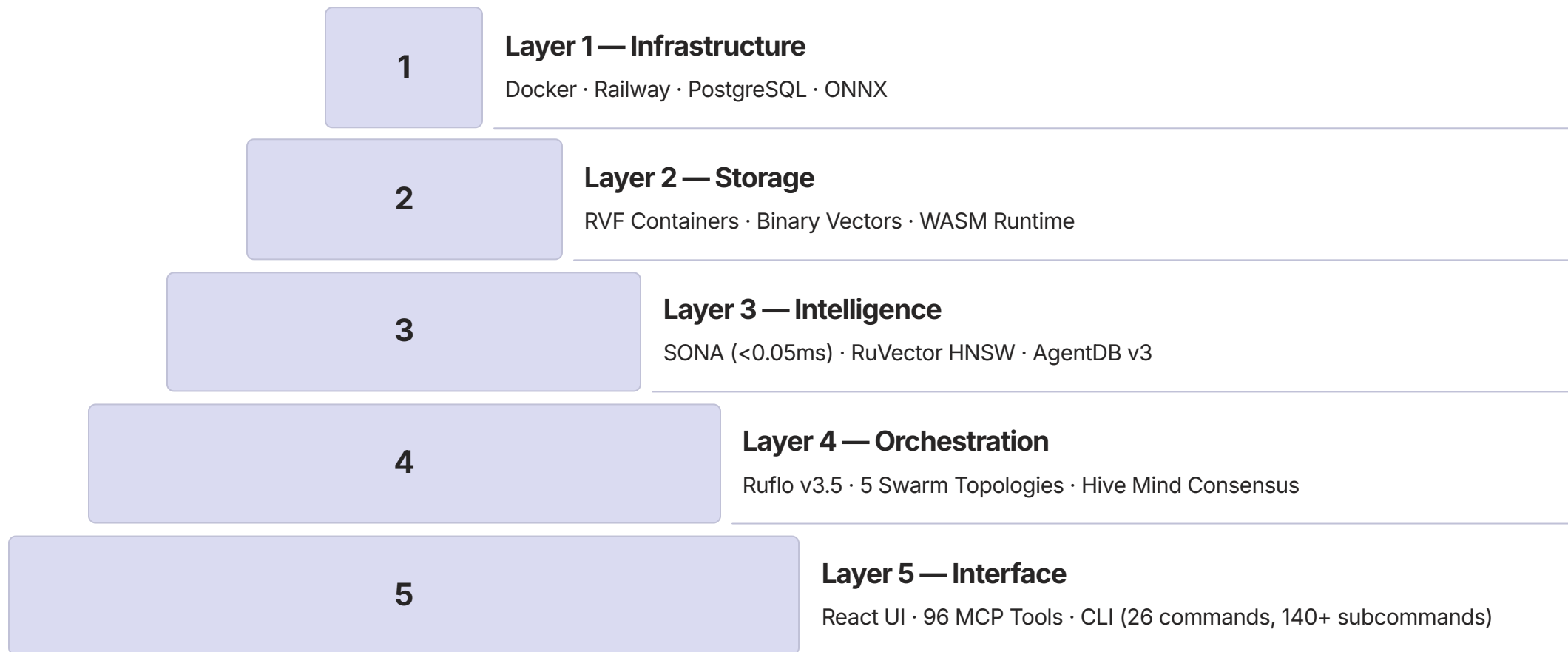
TECHNICAL BRIEFING

V3.5



Five-Layer Architecture Stack

Ruflo v3.5 is built on a coherent, five-layer stack designed for scalability, intelligence, and performance — from infrastructure primitives to user-facing interfaces.



Agent System — 60+ Specialized Types

Ruflo v3.5 ships with over 60 purpose-built agent types, organized into functional families for full software development lifecycle coverage.

Core

- coder, reviewer
- tester, planner
- researcher

Security

- security-architect
- security-auditor

Swarm

- hierarchical-coordinator
- mesh-coordinator
- adaptive-coordinator

GitHub

- pr-manager
- code-review-swarm
- release-manager

SPARC

- specification
- pseudocode
- architecture
- refinement

Swarm Topologies & Consensus Protocols

5 Swarm Topologies

Hierarchical

Queen-led command structure.
Ideal for directed workflows.

Hierarchical-Mesh

Recommended for 10+ agents. Balanced authority with lateral communication.

Star

Broadcast pattern. Central hub distributes tasks to all nodes.

Mesh

Peer-to-peer coordination.
Maximum resilience and redundancy.

Ring

Pipeline topology. Deterministic sequential processing.

4 Consensus Protocols

Byzantine BFT

Fault tolerance up to $f < n/3$. For adversarial environments.

Gossip

Eventual consistency. Highly scalable epidemic propagation.

Raft

Leader-based consensus.
Strong consistency guarantees.

CRDT

Conflict-free replicated data types. No coordination required.



RuVector — High-Performance Vector Engine

RuVector is a drop-in pgvector replacement engineered for production-scale similarity search, with hardware-level acceleration and cross-repository schema isolation.

290+

SQL Vector Functions

Comprehensive function library
for vector operations

12,5...

Max HNSW Speedup

vs. brute-force at scale

<1.2...

Search Latency

On 1 million vectors



SIMD Acceleration

AVX-512, AVX2, and NEON support for maximum hardware utilization across x86 and ARM.



Schema Isolation

Cross-repository schema isolation ensures clean tenant separation without performance penalty.



Drop-In Replacement

Fully compatible pgvector API surface — migrate with zero code changes.

Intelligence Pipeline — SONA & Neural Learning

The SONA intelligence layer implements a continuous four-step learning loop, enabling agents to adapt from experience in near-real-time without catastrophic forgetting.



Flash Attention

2.49x–7.47x speedup on long-context inference

MoE Routing

Mixture-of-Experts for specialized, low-overhead task routing

12 Background Workers

Continuous learning without interrupting active agent workflows

SONA Latency

<0.05ms adaptation response — imperceptible overhead

AIMDS — AI Defense Middleware

AIMDS is a three-layer security pipeline that wraps every AI interaction with validated, chaos-aware threat detection — requiring only a single line of integration code.

1

Inbound Scan

Zod schema validation, path traversal prevention, prompt injection blocking

2

Processing

SQL injection protection, command injection guards, 25-level meta-learning threat classification

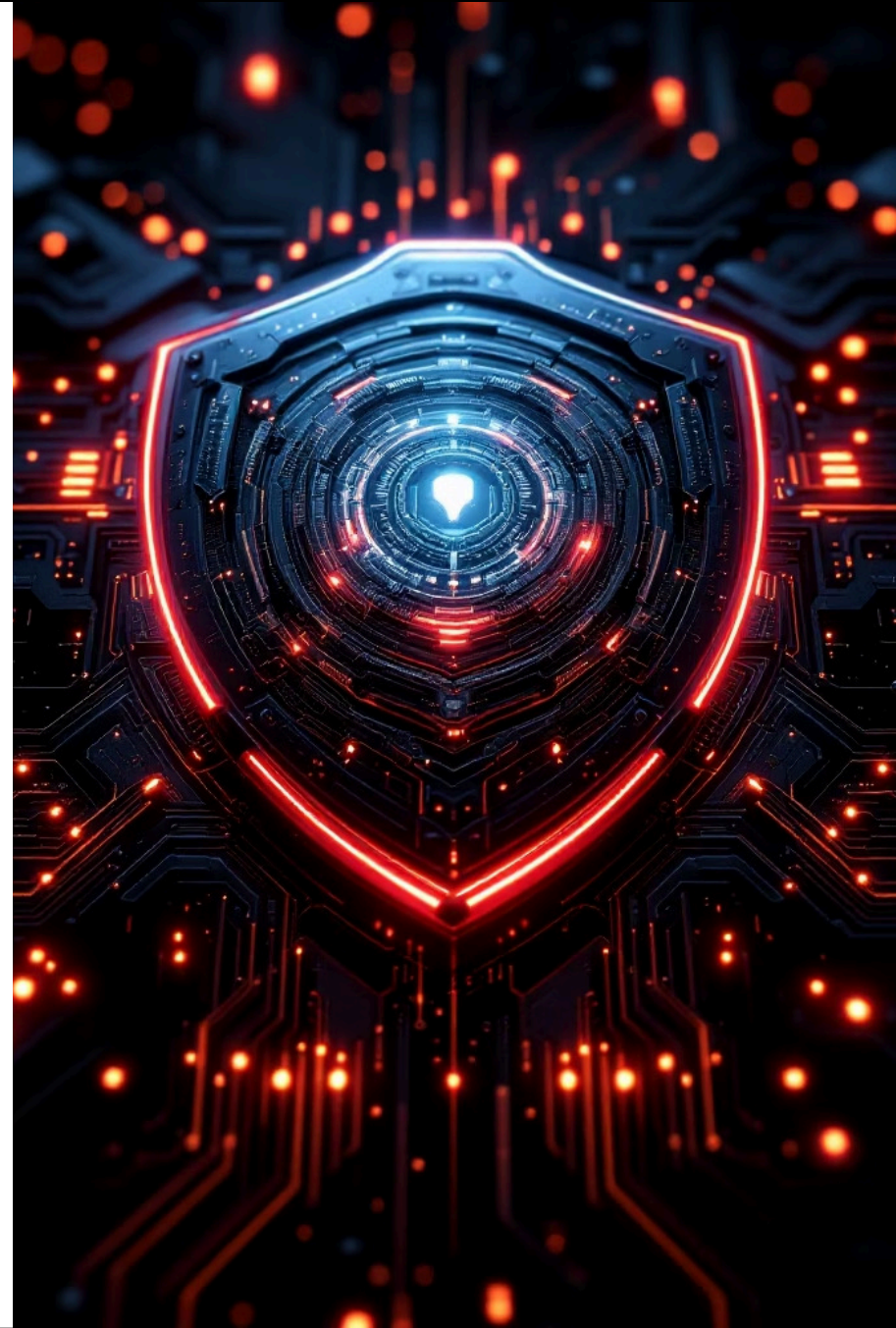
3

Outbound Scan

Lyapunov chaos detection, output sanitization, response integrity verification



One-line integration: `createAIDefence({ enableLearning: true })` — defense activates immediately with adaptive learning enabled.



RVF — Cognitive Container Format

24 Segment Types

RVF packages models, vectors, configuration, and runtime logic into a single portable artifact — self-booting in any browser via a 5.5KB WASM bootstrapper.

VECTOR_SEG

Embedded vector data
with binary encoding

HNSW_SEG

Pre-built HNSW index for
instant search

WASM_SEG

Executable WASM
runtime module

MODEL_SEG

Quantized model weights

Key Capabilities

SQ8 Quantization

79% size reduction —
0.9999 cosine similarity
preserved

Ed25519 Signing

Cryptographic integrity
for every container
artifact

IPFS Distribution

Decentralized, content-
addressed delivery at
global scale

Browser-Native

Zero backend
dependency — full
capability in-browser via
WASM

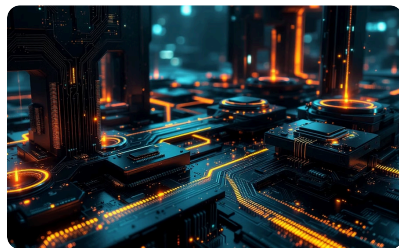
Ask-RuvNet — Reference Implementation

Ask-RuvNet demonstrates the full Ruflo stack in production: a 103,755-entry knowledge base delivered as a 32.5MB browser-native RVF container, with no external database required.



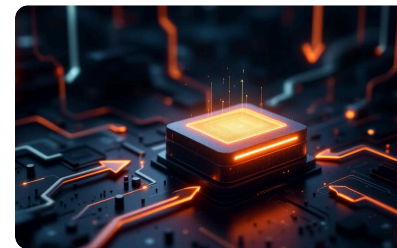
Knowledge Base

103,755 entries — 323 gold + 103,432 architecture entries. RuvectorStore uses binary HNSW with no external DB dependency.



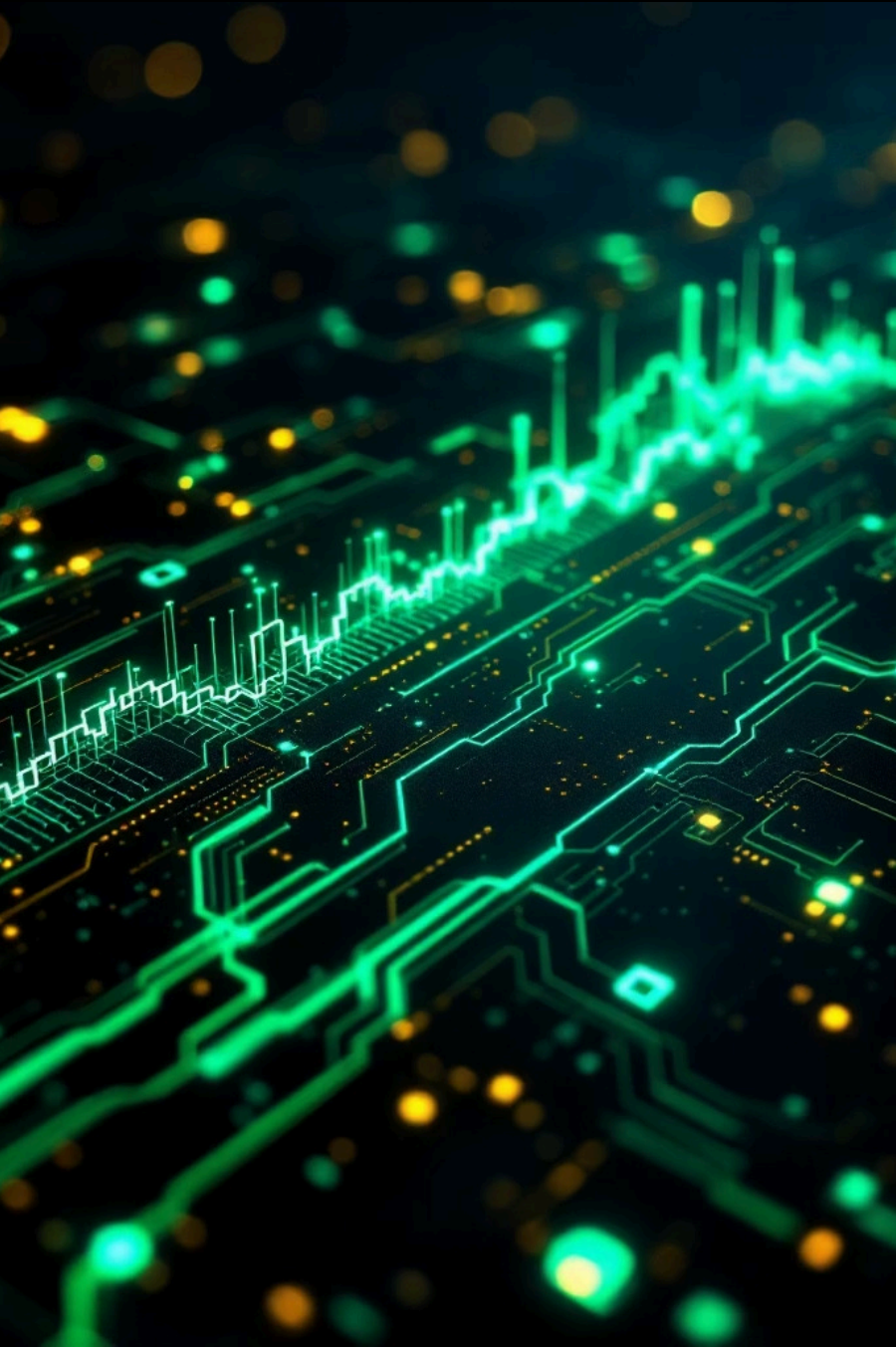
RAG Pipeline

QueryExpander → HybridSearch (BM25 + semantic) → ReRanker → ContextCompressor → MultiHopRetriever. Five-provider LLM fallback chain.



RVF Delivery

Full cognitive container runs in-browser. **32.5MB transfer** — 79% reduction via SQ8 quantization. Zero backend infrastructure needed.



Performance & Benchmarks

Ruflo v3.5 delivers measurable, independently verifiable performance gains across every subsystem — from sub-millisecond agent adaptation to browser-native vector search.

7.47x

Flash Attention Peak

Speedup on long-context inference windows

12,500x

HNSW Search Peak

Faster than brute-force similarity search

75%

Memory Reduction

Via scalar quantization at full deployment

MCP Response

<100ms

CLI Startup

<500ms

SONA Adapt

<0.05ms

Browser WASM

<200ms on 100K vectors

Hooks & Background Workers

27 Lifecycle Hooks

Hooks provide deep integration into every stage of agent activity, enabling custom instrumentation, routing, and learning triggers.

01

Edit Hooks

pre/post-edit — intercept and audit all file mutations

02

Command Hooks

pre/post-command — wrap CLI execution with custom logic

03

Intelligence Hooks

pretrain, intelligence, route, explain — shape learning and routing behavior

04

Coverage Hooks

coverage-route, coverage-suggest, coverage-gaps — test coverage orchestration

12 Background Workers

Workers run continuously in the daemon, handling learning, optimization, and documentation without blocking active workflows.

ultralearn

Continuous model weight updates from agent experience

optimize

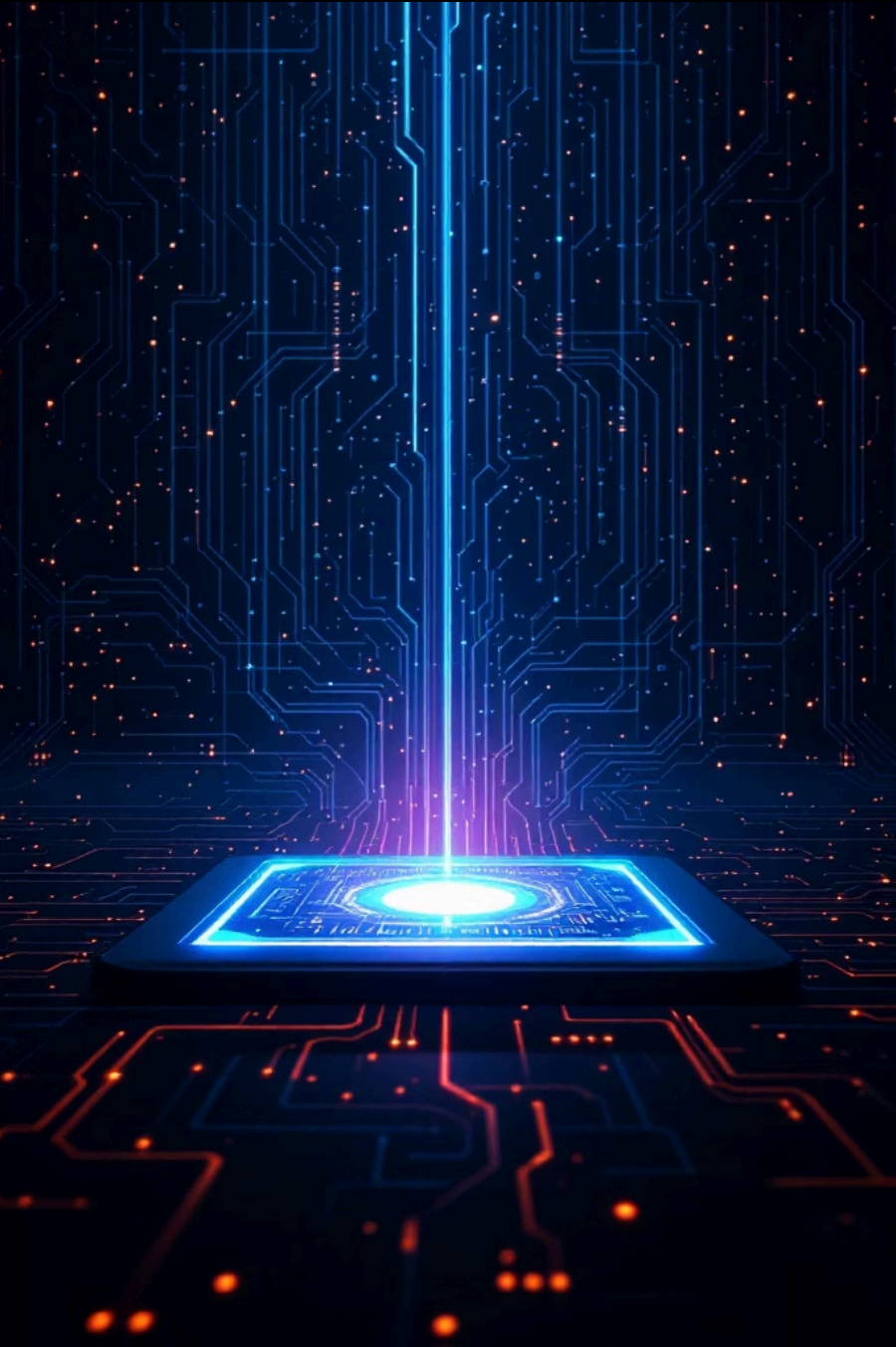
Runtime performance tuning and resource rebalancing

consolidate

EWC++ memory consolidation to prevent forgetting

benchmark

Continuous performance regression detection



Getting Started & Integration

Ruflo v3.5 is designed for zero-friction adoption — from a single init command to full multi-provider, multi-IDE deployment in minutes.



Quick Start

```
npx ruflo@latest init --wizard
```

```
Daemon: npx ruflo@latest daemon start
```

```
Shell completions: bash, zsh, fish, powershell
```



MCP & LLM Providers

96 MCP tools via stdio transport. Providers: OpenAI, Groq, Anthropic, OpenRouter, DeepSeek — with automatic fallback chain.



IDE Support

Native integration with Claude Code, VS Code, and JetBrains. Full IntelliSense and agent-aware tooling out of the box.



GitHub: github.com/ruvnet/ruflo — full documentation, examples, and community resources.